

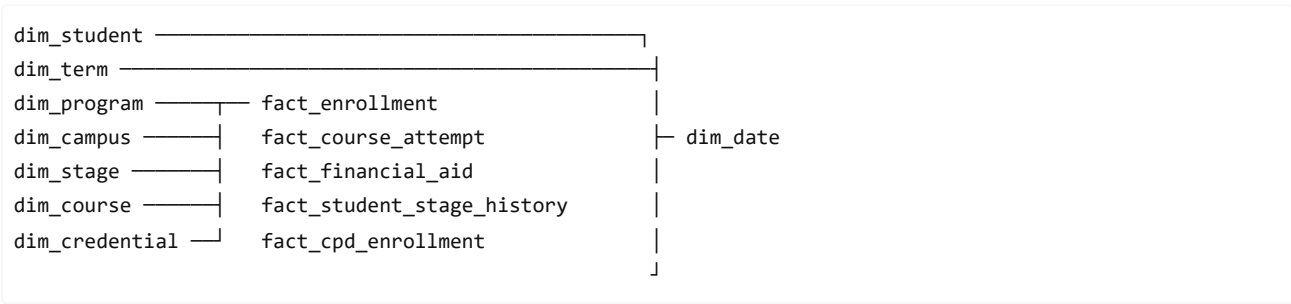
San Jacinto College — Dimensional Data Model

Version: 0.1

Purpose: Defines the star schema to support all 8 Cortex demo use cases. Used as the specification for synthetic data generation.

Model Overview

The model uses a **star schema** with 5 fact tables and 8 dimension tables. The grain varies by fact table (see each table). All facts join to `dim_student` and `dim_term` at minimum.



Dimension Tables

`dim_student`

Grain: One row per student

~32,000 rows (credit students; CPD-only students handled separately in `dim_cpd_student`)

Column	Type	Description	Generation Rules
<code>student_id</code>	VARCHAR PK	Unique student identifier (e.g. S000001)	Sequential, zero-padded
<code>first_name</code>	VARCHAR	First name	Realistic names weighted by ethnicity
<code>last_name</code>	VARCHAR	Last name	Realistic names weighted by ethnicity
<code>gender</code>	VARCHAR	Gender identity	59% Female, 41% Male
<code>ethnicity</code>	VARCHAR	Race/ethnicity	See distribution below
<code>age_at_first_enrollment</code>	INT	Age when first enrolled	See distribution below
<code>dob</code>	DATE	Date of birth	Derived from age + first enrollment term
<code>zip_code</code>	VARCHAR	Home zip code	97% Texas — weighted to SJC service area zips (see brand doc)
<code>state</code>	VARCHAR	State of residence	97% TX, 3% other
<code>pell_eligible</code>	BOOLEAN	Receives Federal Pell Grant	56% TRUE
<code>first_gen_college</code>	BOOLEAN	First-generation college student	~45% TRUE (correlated with Hispanic/low-income)
<code>high_school_district</code>	VARCHAR	Feeder ISD	Channelview, Clear Creek, Crosby, Deer Park, etc.
<code>dual_credit</code>	BOOLEAN	Started as dual credit student	~10% TRUE

hs_gpa	DECIMAL(3,2)	High school GPA	Normal distribution, mean 2.8, SD 0.5, range 1.5–4.0
tsi_math_exempt	BOOLEAN	TSI math exempt/satisfied	~ 65% TRUE
tsi_reading_exempt	BOOLEAN	TSI reading/writing exempt	~ 70% TRUE
meningitis_submitted	BOOLEAN	Meningitis doc on file	~ 85% TRUE of admitted+ students
nso_complete	BOOLEAN	New Student Orientation complete	~ 80% TRUE of admitted+ students
created_date	DATE	Record creation date	Varies by term cohort

Ethnicity Distribution:

Value	%
Hispanic or Latino	63.3%
White	15.2%
Black or African American	8.5%
Asian	5.0%
Two or More Races	1.9%
Unknown / Not Reported	5.1%
American Indian / Alaska Native	0.6%
Native Hawaiian / Pacific Islander	0.4%

Age Distribution (at first enrollment):

Range	%
Under 18 (dual credit)	5%
18–21 (traditional)	45%
22–25	20%
26–35	18%
36–50	10%
51+	2%

dim_term

Grain: One row per academic term

Column	Type	Description	Generation Rules
term_id	VARCHAR PK	e.g. 2024FA, 2025SP, 2025SU	
term_name	VARCHAR	e.g. "Fall 2024"	
term_season	VARCHAR	Fall / Spring / Summer	

academic_year	VARCHAR	e.g. "2024–25"	
start_date	DATE	First day of term	
end_date	DATE	Last day of term	
census_date	DATE	Official enrollment count date	~12th class day
is_current_term	BOOLEAN	Flag for current term	Only one TRUE at a time

Generate: 6 terms minimum (3 academic years: 2022–23, 2023–24, 2024–25) to support trend queries.

dim_program

Grain: One row per program (credit credential)

~200+ rows

Column	Type	Description	Generation Rules
program_id	VARCHAR PK	e.g. BUS-AAS-001	
program_name	VARCHAR	Full program name	From programs-catalog.md
credential_type	VARCHAR	AAS / AA / AS / Certificate of Technology / Level 2 Cert / Occupational Cert / ATC / BAT / BAS / AAT	
area_of_study	VARCHAR	Top-level academic area	See values below
is_transfer	BOOLEAN	Academic transfer pathway (AA/AS/AAT)	
is_workforce	BOOLEAN	Workforce/technical credential	
is_critical_field	BOOLEAN	THECB critical field designation	~30% TRUE (Health Sciences, STEM, trades)
sch_required	INT	Total SCH required for completion	AA/AS: 60; AAS: 60–72; Certs: 15–45
typical_completion_semesters	INT	Expected semesters to complete full-time	AA/AS: 4; AAS: 4–6; Certs: 2–4
licensure_required	BOOLEAN	Requires licensure/certification exam	~15% TRUE (Nursing, Dental, EMS, etc.)
campus_primary	VARCHAR	Primary campus where program is offered	Central / North / South / Gen Park / Maritime

Area of Study Values:

- Arts, Humanities, Communications & Design
- Business
- Construction, Industry, Manufacturing & Transportation
- Education
- Health Sciences
- Information Technology
- Science & Mathematics
- Social & Behavioral Sciences

- Public Safety & Criminal Justice

dim_course

Grain: One row per unique course (prefix + number)

~800–1,000 rows

Column	Type	Description	Generation Rules
course_id	VARCHAR PK	e.g. MATH-1314	
course_prefix	VARCHAR	e.g. MATH, ENGL, BIOL	
course_number	VARCHAR	e.g. 1314	
course_title	VARCHAR	Full title	e.g. "College Algebra"
sch_value	INT	Semester credit hours	Mostly 3; some 1–4
course_level	VARCHAR	College-level / Developmental	~15% Developmental
delivery_mode	VARCHAR	Face-to-face / Online / Hybrid	See distribution below
area_of_study	VARCHAR	Matches dim_program area	
is_gateway	BOOLEAN	First college-level math or reading/writing	~2% TRUE — important for success points

Delivery Mode Distribution (per section):

Mode	%
Face-to-face	38%
Online	30%
Hybrid	32%

dim_campus

Grain: One row per campus

5 rows

campus_id	campus_name
CEN	Central Campus
NOR	North Campus
SOU	South Campus
GEN	Generation Park Campus
MAR	Maritime Campus

dim_stage

Grain: One row per TargetX student stage

11 rows

stage_id	stage_name	flow_type	is_active_student	sort_order
1	Prospective	Record-Triggered	FALSE	1
2	Applicant	Record-Triggered	FALSE	2
3	Not Interested	Record-Triggered	FALSE	99
4	Admitted	Record-Triggered	FALSE	3
5	Eligible to Register – New	Record-Triggered	FALSE	4
6	Register Pending	Record-Triggered	FALSE	5
7	Eligible to Register – Continuing	Date-Based	FALSE	6
8	Enrolled	Record-Triggered	TRUE	7
9	Graduate / Alumni	Record-Triggered	FALSE	8
10	Stop-Out	Date-Based	FALSE	9
11	Recruit Backs	Date-Based	FALSE	10

dim_credential

Grain: One row per awarded credential type

Reference table

Column	Type	Description
credential_id	VARCHAR PK	
credential_type	VARCHAR	AAS, AA, AS, AAT, BAT, BAS, Cert of Tech, Level 2 Cert, Occupational Cert, ATC
credential_level	VARCHAR	Bachelor / Associate / Certificate
transfer_eligible	BOOLEAN	TRUE for AA, AS, AAT

Fact Tables

fact_enrollment

Grain: One row per student per term

~85,000 rows (32,000 students × ~2.7 active terms on average)

Column	Type	Description	Generation Rules
enrollment_id	BIGINT PK	Surrogate key	
student_id	VARCHAR FK	→ dim_student	
term_id	VARCHAR FK	→ dim_term	
program_id	VARCHAR FK	→ dim_program (declared major)	

campus_id	VARCHAR FK	→ dim_campus	
stage_id	INT FK	→ dim_stage (stage at census date)	
enrollment_type	VARCHAR	First-Time-in-College / Transfer / Continuing / Re-admit	FTIC: 25%, Transfer: 10%, Continuing: 60%, Re-admit: 5%
total_sch_attempted	INT	Total SCH registered	Full-time (12+): 33.8% ; Part-time (<12): 66.2%
total_sch_completed	INT	SCH successfully completed (grade ≥ D or pass)	
sch_on_pathway	INT	SCH that count toward declared program	~85% of completed SCH on average
is_online_only	BOOLEAN	All courses online	30% TRUE
has_some_online	BOOLEAN	Mix of online and face-to-face	32% TRUE among non-online-only
payment_status	VARCHAR	Paid / Payment Plan / Aid Authorized / Unpaid	Unpaid → Register Pending stage
mf_hold_active	BOOLEAN	Active MF hold blocking registration	Admitted stage only
is_first_term	BOOLEAN	First ever term at SJC	
withdrew_term	BOOLEAN	Withdrew from all courses mid-term	~8% TRUE
financial_aid_term	BOOLEAN	Received any financial aid this term	~60% TRUE

SCH Distribution:

Range	Label	% of enrollments
1–5	Very Part-Time	15%
6–8	Part-Time	28%
9–11	Near Full-Time	24%
12–14	Full-Time	26%
15+	Overload	7%

Stage at Census Date Distribution (current term):

Stage	%
Enrolled	62%
Register Pending	4%
Eligible to Register – Continuing	8%
Eligible to Register – New	4%
Admitted	3%
Graduate / Alumni	8%

Stop-Out	6%
Recruit Backs	2%
Prospective / Applicant	2%
Not Interested	1%

fact_course_attempt

Grain: One row per student per course section per term

~250,000 rows (average ~3 courses per enrolled student per term)

Column	Type	Description	Generation Rules
attempt_id	BIGINT PK	Surrogate key	
student_id	VARCHAR FK	→ dim_student	
term_id	VARCHAR FK	→ dim_term	
course_id	VARCHAR FK	→ dim_course	
campus_id	VARCHAR FK	→ dim_campus	
section_id	VARCHAR	Course section identifier	
delivery_mode	VARCHAR	Face-to-face / Online / Hybrid	
instructor_id	VARCHAR	Anonymized instructor ID	
grade	VARCHAR	A / B / C / D / F / W / I	See distribution below
grade_points	DECIMAL(3,1)	4.0 scale	A=4, B=3, C=2, D=1, F=0, W/I=null
ac_success	BOOLEAN	Grade of A, B, or C	~72% TRUE overall
withdrew	BOOLEAN	W grade (student-initiated withdrawal)	~12% TRUE
sch_earned	INT	SCH earned (0 if W/F)	
is_developmental	BOOLEAN	Dev ed course	~15% of all attempts
attendance_pct	DECIMAL(5,2)	% of class sessions attended	Normal dist; mean 82%, SD 15%
midterm_grade	VARCHAR	A / B / C / D / F	Leading indicator for early alert

Grade Distribution (non-withdrawn):

Grade	% of non-W attempts
A	35%
B	22%
C	15%
D	8%
F	20%

Restrictions:

- Withdrawal rate higher for: developmental courses (~**22% W**), first-term students (**+5%**), part-time working students
- A–C success rate higher for: full-time students, continuing students, students with complete financial aid
- A–C success rate lower for: developmental courses (~**55%**), students with attendance_pct < 70%

fact_financial_aid

Grain: One row per student per term per aid type

~60,000 rows

Column	Type	Description	Generation Rules
aid_id	BIGINT PK	Surrogate key	
student_id	VARCHAR FK	→ dim_student	
term_id	VARCHAR FK	→ dim_term	
aid_type	VARCHAR	Pell Grant / TPEG / Workforce Solutions / Institutional / Loan / Work-Study	
aid_amount_awarded	DECIMAL(8,2)	Aid awarded for the term	
aid_amount_disbursed	DECIMAL(8,2)	Aid actually disbursed	
unmet_need	DECIMAL(8,2)	Cost of attendance minus all aid	
disbursement_status	VARCHAR	Pending / Disbursed / Cancelled	
satisfactory_academic_progress	BOOLEAN	SAP met	~85% TRUE
enrollment_intensity_required	INT	Minimum SCH for aid	Pell = 6 SCH minimum for half
cumulative_loan_balance	DECIMAL(10,2)	Total loan debt to date	Only for loan-type records

Aid Type Distribution (among aid recipients):

Type	% of aided students
Pell Grant	56% of all students; avg award ~\$3,200/term
TPEG	15% of Pell-eligible students
Institutional Grant	10%
Workforce Solutions	8% of CPD/workforce program students
Student Loan	12%
Work-Study	3%

Unmet Need Distribution:

Range	%
\$0 (fully covered)	20%
\$1–\$500	25%
\$501–\$1,500	30%

\$1,501–\$3,000	15%
\$3,001+	10%

fact_student_stage_history

Grain: One row per student per stage transition

~90,000 rows (avg ~2.8 stage changes per student lifetime)

Column	Type	Description	Generation Rules
history_id	BIGINT PK	Surrogate key	
student_id	VARCHAR FK	→ dim_student	
stage_id	INT FK	→ dim_stage	
stage_start_date	DATE	Date entered this stage	
stage_end_date	DATE	Date exited this stage (NULL if current)	
days_in_stage	INT	Computed duration	
previous_stage_id	INT	Stage they came from	
trigger_type	VARCHAR	Record-Triggered / Date-Based	
term_id	VARCHAR FK	Term associated with transition	

Stage Transition Rules:

- Prospective → Applicant: median 14 days after inquiry
- Applicant → Admitted: median 21 days after application accepted
- Admitted → Eligible to Register – New: MF hold cleared, median 7 days
- Eligible to Register – New → Register Pending: median 5 days
- Register Pending → Enrolled: payment confirmed, median 3 days
- Enrolled → Eligible to Register – Continuing: triggered 4 weeks before term end
- Eligible to Register – Continuing → Stop-Out: after 3 terms without re-enrollment
- Enrolled → Graduate/Alumni: upon credential award
- **~8% of Applicants → Not Interested** (Banner = "Student Withdrawal")
- **~15% of Eligible to Register – New → Recruit Backs** (stalled > 6 months)

fact_cpd_enrollment

Grain: One row per CPD student per course enrollment

~20,000 rows

Column	Type	Description	Generation Rules
cpd_enrollment_id	BIGINT PK	Surrogate key	
cpd_student_id	VARCHAR FK	→ dim_cpd_student (separate dimension)	
term_id	VARCHAR FK	→ dim_term	
course_id	VARCHAR	CPD course code	

course_name	VARCHAR	CPD course name	
cpd_center	VARCHAR	EDGE / BIT / Corporate / Community & Health	See distribution below
campus_id	VARCHAR FK	→ dim_campus	
delivery_mode	VARCHAR	In-Person / Online / Hybrid	
enrollment_date	DATE	Registration date	
completion_status	VARCHAR	Completed / In Progress / Dropped / No-Show	Completed: 78%, Dropped: 12%, No-Show: 10%
seat_capacity	INT	Max seats in course	
grant_funded	BOOLEAN	TPEG or Workforce Solutions grant	~25% TRUE
employer_sponsored	BOOLEAN	Paid by employer / corporate contract	~20% TRUE
linked_to_credit	BOOLEAN	"Linked" course — credit applicable	~10% TRUE

CPD Center Distribution:

Center	%
EDGE (Aerospace/Aviation/Mfg)	18%
BIT (IT/AI/Business/Logistics)	25%
Corporate (Apprenticeships/Grants)	22%
Community & Health	35%

Key Metric Definitions (for validation)

Use these to validate generated data produces correct aggregate values:

Metric	Target Value	Calculation
Total active credit students	~31,812	COUNT(DISTINCT student_id) in current term enrolled
Full-time rate	33.8%	% with total_sch_attempted ≥ 12
Online-only rate	30%	% with is_online_only = TRUE
Pell eligibility rate	56%	% of dim_student with pell_eligible = TRUE
Hispanic enrollment	63.3%	% of dim_student with ethnicity = 'Hispanic or Latino'
Female enrollment	59%	% of dim_student with gender = 'Female'
A–C course success rate	~72%	% of fact_course_attempt where ac_success = TRUE (excl. W)
Withdrawal rate	~12%	% of fact_course_attempt where withdrew = TRUE
Fall-to-Fall persistence	~55%	% of Fall enrolled students also enrolled next Fall
Fall-to-Spring persistence	~68%	% of Fall enrolled students also enrolled same Spring

Graduation rate (3-yr FTIC)	~18%	% of FTIC cohort awarding credential within 3 years
Transfer rate	~22%	% of AA/AS/AAT completers who transfer
Stop-Out rate	~6%	% of student population in Stop-Out stage
Recruit Backs rate	~2%	% of student population in Recruit Backs stage

Referential Integrity & Constraint Notes

- Every `fact_enrollment` row must have a valid `student_id`, `term_id`, `program_id`, `campus_id`
- A student in stage **Enrolled** must have `payment_status` IN ('Paid', 'Payment Plan', 'Aid Authorized')
- A student in stage **Register Pending** must have `payment_status` = 'Unpaid' and a future/current academic profile
- A student in stage **Admitted** must have `mf_hold_active` = TRUE
- A student in stage **Eligible to Register – New** must have all three: `tsi*_exempt` = TRUE, `meningitis_submitted` = TRUE, `nso_complete` = TRUE, and `mf_hold_active` = FALSE
- `sch_on_pathway` must be ≤ `total_sch_completed`
- `aid_amount_disbursed` must be ≤ `aid_amount_awarded`
- Students with `pell_eligible` = FALSE should not have Pell Grant records in `fact_financial_aid`
- `dual_credit` = TRUE students should have `age_at_first_enrollment` < 18
- Developmental course attempts (`is_developmental` = TRUE) should only appear in first 1–2 terms

Correlated Distributions (important for realism)

These correlations should be applied during generation — not all fields are independent:

If...	Then...
<code>pell_eligible</code> = TRUE	Higher probability of <code>unmet_need</code> > 0, <code>first_gen</code> = TRUE, part-time enrollment
<code>ethnicity</code> = Hispanic	Higher probability of <code>pell_eligible</code> = TRUE (65%), <code>first_gen</code> = TRUE (50%)
<code>is_online_only</code> = TRUE	Higher probability of part-time, age > 25, working student
<code>enrollment_type</code> = FTIC	Higher probability of dev ed courses, lower A–C success rate in first term
<code>attendance_pct</code> < 70%	Higher probability of W grade or F grade
<code>total_sch_attempted</code> ≥ 12	Higher A–C success rate, higher persistence rate
<code>financial_aid_term</code> = TRUE	Lower withdrawal rate, higher persistence
<code>is_developmental</code> = TRUE (course)	Lower A–C success rate (55%), higher W rate (22%)
<code>program.licensure_required</code> = TRUE	Higher course success required; lower graduation rate if failed
<code>stage</code> = Stop-Out	<code>terms_since_last_enrollment</code> between 3–8
<code>stage</code> = Recruit Backs	Time in Eligible to Register – New > 180 days, no registration ever